Harish Venkatesan M R

Robotics Engineer | ROS 2 | Autonomous Systems | Computer Vision | LLM Robotics

York, UK (open to relocation) | 07407 307096 | mrharishv29@gmail.com

LinkedIn | GitHub | Portfolio

PERSONAL PROFILE

Robotics engineer with an MSc in Intelligent Robotics from the University of York and 18 months of commercial software engineering experience. Builds autonomous robots end to end, from sensors and embedded control on Arduino and Raspberry Pi to ROS 2 perception, SLAM and Nav2 navigation in Gazebo, including a natural-language robot that turns spoken and typed commands into safe, validated actions using an LLM. Led a 4-person team to deliver a rescue robot with a custom YOLOv5 model achieving 86% mAP. Seeking a Robotics, Automation or Controls Engineer role in the UK.

KEY SKILLS

- **Robotics Software:** ROS 2 (Jazzy), Nav2, SLAM Toolbox, Gazebo Harmonic, rclpy, URDF/xacro, RViz
- **Autonomy and Control:** SLAM, path planning, sensor fusion, state machines, PID, computed-torque control
- **Vision and AI:** YOLOv5/YOLO11, OpenCV, PyTorch, LLM integration (Ollama, Claude API), speech-to-text
- **Embedded and Sensors:** Arduino, Raspberry Pi, UART, I2C, LiDAR, IR/ultrasonic, encoders, cameras
- **Programming and Tools:** Python, C/C++, Java, MATLAB/Simulink, Git, Linux, Fusion 360, Inventor

ROBOTICS PROJECTS

LLM Command Robot on ROS 2 Personal project, 2026

Python, ROS 2 Jazzy, Gazebo Harmonic, Nav2, SLAM Toolbox, Ollama/Claude LLM, YOLO11, OpenCV, faster-whisper

- Built a home-assistant robot in a simulated 7-room house that executes natural-language commands such as "go to the kitchen", "where is the cup?" or "follow me", spoken or typed.
- Designed a hybrid command pipeline: a fast rule-based parser handles common commands and an LLM (local Ollama or Claude API) handles free-form requests, with every LLM output schema-validated before the robot acts.
- Implemented room-to-room navigation over a place graph with Dijkstra routing through doorways, plus Nav2 and SLAM Toolbox integration switchable from a single launch argument.
- Added camera perception (YOLO11 plus colour detection) with a semantic object memory for finding and patrolling, LiDAR-based follow-me that filters out known walls and furniture, and voice control via faster-whisper with spoken replies.
- Visualised remembered objects, the tracked person and detections in RViz, and validated the system with 112 passing unit tests and headless simulation runs of multi-room tours.

6-DOF Robot Arm Control: Computed-Torque vs PID (MATLAB/Simulink) Personal project, 2025

MATLAB, Simulink, Robotics System Toolbox

- Modelled a UR5 6-DOF arm's forward and inverse kinematics (DH parameters) and rigid-body dynamics, and generated quintic joint trajectories for a pick-and-place task.
- Built PID and computed-torque controllers in Simulink, using the mass matrix, Coriolis and gravity terms for model-based control.
- Ran a robustness study with an unmodelled 2 kg payload, 10% mass error and sensor noise: computed-torque control cut RMS end-effector tracking error by about 70% versus PID and peak error from 49 mm to 15 mm.
- Identified a noise trade-off, with computed-torque amplifying sensor noise into torque jitter more than PID, and documented results and limitations in a reproducible GitHub repo.

Autonomous Search and Rescue Robot University of York, 2024

Team Lead (team of 4) - Python, C++, Arduino, Raspberry Pi, YOLOv5, OpenCV

- Led a team of 4 engineers, assigning subsystem ownership across ground and aerial platforms and integrating all workstreams into a single working system for disaster response.
- Delivered autonomous maze navigation for a Mecanum-wheeled robot, fusing IR and ultrasonic sensor data for obstacle avoidance, with motor control on Arduino and high-level logic on Raspberry Pi in Python.
- Trained a custom YOLOv5 model to detect victims and obstacles, achieving 86% mAP, and fed live detections directly into navigation decisions.
- Developed the serial communication layer linking the sensor array to a custom two-jaw gripper, enabling the robot to grasp a rescue object and return it to the start position without operator input.

PROFESSIONAL EXPERIENCE

Associate Engineer - Technology, Virtusa Consulting Services 05/2022 - 09/2023
Chennai, India

- Software test engineer in a 10-15 person Agile/Scrum team delivering enterprise applications built on Java, Spring Boot, REST APIs and React.js, across the full development lifecycle.
- Designed and executed unit and integration tests for backend services, REST APIs and front-end features, improving the reliability of production releases.
- Debugged defects across backend and front end: reproduced issues, isolated root causes and verified fixes before release.
- Carried out peer code reviews focused on correctness, test coverage and maintainability.
- Wrote and maintained test and technical documentation, and supported CI/CD pipelines, AWS deployments and release management.

Intern - Delivery (Team Lead), Virtusa Consulting Services 01/2022 - 03/2022
Chennai, India | 3-month project delivery challenge

- Led a team of 6 interns in a 3-month build challenge; when the team dropped to 3, redistributed the work, kept morale up and still delivered the working build on time.
- Contributed to Java/Spring Boot development, testing and code reviews; converted to a full-time role on graduation.

Robotics Team Member, SP Robotics 03/2018 - 08/2018

- Designed small, simple hands-on tasks that built students' robotics skills step by step, from basic components to working robots.
- Prepared kits and equipment for each session and adapted explanations to different ability levels, working alongside senior staff.

EDUCATION

MSc Intelligent Robotics, University of York 09/2023 - 01/2025

- Modules: Robot Kinematics and Dynamics; Control and Instrumentation for Robotics; Practical Robotics; Embedded Systems Design and Programming; Machine Learning

BE Computer Science and Engineering, KPR Institute of Engineering and Technology, India 08/2018 - 05/2022

- First Class. Modules: Data Structures; Object-Oriented Programming; Computer Networks; Human-Computer Interaction; Natural Language Processing

ACHIEVEMENTS AND CERTIFICATIONS

- Grand Finalist, Virtusa NeuralHack Season 5 (2021): top 250 of 100,000+ participants in a national hackathon; earned a pre-placement offer.
- Oracle Certified Associate, Java SE 8 Programmer - Oracle (2022)